

AI/ML Intern Assessment Report

Support Ticket Categorization using NLP and Machine Learning

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GitHub Repository: <https://github.com/Mohammed-Najmudeen-K/ticket-categorizer>

Project Overview

The objective of this project is to develop an AI-powered support ticket categorization system that automatically classifies customer support tickets into predefined categories.

The application processes the text of incoming support tickets, converts the text into numerical features using TF-IDF vectorization, and predicts the appropriate category using a Logistic Regression classifier.

The system is designed to assist customer support teams by reducing manual ticket routing and improving response efficiency.

Problem Statement

Support teams receive hundreds of customer requests every day.

Manually assigning each ticket to the correct department is time-consuming and prone to errors.

This project automates ticket classification into the following categories:

- Billing
 - Technical
 - HR
 - General
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Technologies Used

Technology	Purpose
Python	Programming Language
Pandas	Data Processing

Technology	Purpose
Scikit-learn	Machine Learning
NLTK	Text Preprocessing
TF-IDF	Feature Extraction
Logistic Regression Classification	
Streamlit	Web Application
Joblib	Model Serialization

Dataset

A balanced dataset containing support tickets across four categories was created.

Categories included:

- Billing
- Technical
- HR
- General

Each ticket contains:

- Subject
- Description
- Category

The dataset was preprocessed before training.

Machine Learning Pipeline

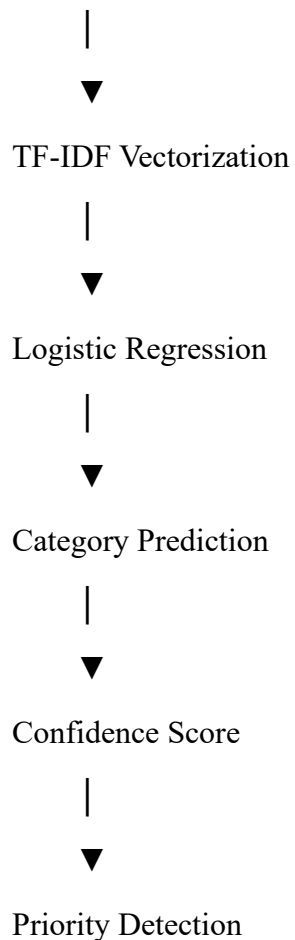
Support Ticket

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Text Cleaning

(Lowercase, Remove Punctuation,
Stopwords, Lemmatization)



Text Preprocessing

The following preprocessing techniques were applied:

- Convert text to lowercase
- Remove punctuation
- Remove numbers
- Remove stop words
- Lemmatize words
- Remove extra spaces

These steps improve the quality of the text before feature extraction.

Feature Engineering

The project uses TF-IDF (Term Frequency–Inverse Document Frequency) to convert textual data into numerical vectors.

Advantages:

- Captures important words
 - Reduces influence of common words
 - Efficient for text classification
 - Lightweight and fast
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Model Selection

The final model uses Logistic Regression.

Reasons for selection:

- Performs well on text classification
 - Fast training and prediction
 - Produces probability scores
 - Easy to interpret
 - Suitable for small and medium datasets
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Evaluation

The model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

The trained model achieved strong classification performance on the generated dataset.

Streamlit Application Features

The application includes:

- Real-time ticket prediction
- Confidence score
- Human review threshold

- Priority detection
- Prediction history
- CSV download option

Sample Predictions

Input Ticket	Predicted Category	Confidence
My bank account was debited twice after making the payment.	Billing	80.57%
I need my experience certificate for my new job.	HR	59.21%
The mobile application crashes immediately after opening.	Technical	67.64%
Where is your company headquarters located?	General	72.58%

Bonus Features Implemented

- ✓ Confidence Score
- ✓ Human Review Threshold
- ✓ Priority Detection
- ✓ Streamlit Dashboard
- ✓ Prediction History
- ✓ CSV Export

Future Improvements

With additional time and access to real-world datasets, the following enhancements can be implemented:

- Train on real customer support tickets
- Fine-tune transformer models such as BERT
- Multi-language support
- REST API deployment
- Continuous learning using user feedback

- Cloud deployment using Streamlit Cloud or Docker

Conclusion

This project demonstrates the complete workflow of an NLP-based text classification system, from preprocessing and feature extraction to model training and deployment.

The application provides fast, interpretable predictions with confidence scores and additional decision-support features such as priority tagging and manual review for uncertain predictions. The solution is suitable as a prototype for automated support ticket routing in enterprise environments.

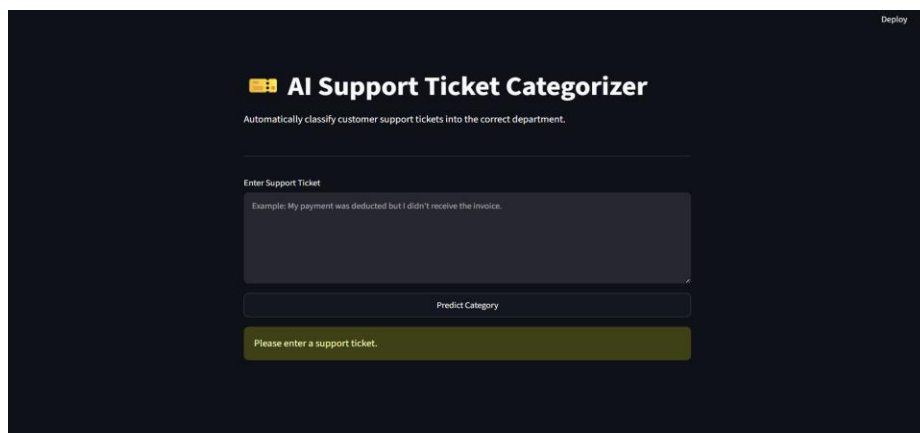
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Screenshots

Dashboard



Prediction Result

